Takyoung Kim

Ph.D. Candidate at the University of Illinois Urbana-Champaign (UIUC)

Thomas M. Siebel Center for Computer Science, 201 North Goodwin Avenue, Urbana, IL 61801

youngerous.github.io | tk30@illinois.edu | in/takyoung | @takyoung_kim | Google Scholar

Interests

My research interest lies in Natural Language Processing, Spoken Language Understanding, and Conversational AI. I am particularly interested in developing companion AI systems that can (1) understand social and cognitive context beyond the textual domain, (2) provide personalized and adaptive assistance, and (3) seamlessly interact within the physical world.

Education

University of Illinois Urbana-Champaign, *Urbana, IL, USA* — Aug 2024 – Present
Doctor of Philosophy, Computer Science
Advisor: Dilek Hakkani-Tür

Korea University, *Seoul, South Korea* — Sep 2020 – Aug 2022
Master of Engineering, Industrial Management Engineering
Advisor: Pilsung Kang (currently affiliated with Seoul National University)

Sungkyunkwan University, *Seoul, South Korea* — Mar 2013 – Aug 2020
Bachelor of Science in Engineering, Interdisciplinary Program of Convergent Software
Bachelor of Media & Communication *(Leave of absence for military service: Feb 2015 - Nov 2016)*

Employment

Meta Superintelligence Labs | Voice Modeling Team *Menlo Park, CA, USA*
Research Scientist Intern (Mentors: Duc Le, Naoyuki Kanda, Abdelrahman Mohamed) — May 2026 – Aug 2026
Worked on data selection for improving speech recognition capabilities of conversational tutor

Amazon Alexa AI | Conversational Assistant Modeling and Learning *Bellevue, WA, USA*
Applied Scientist Intern (Mentors: Jinseok Nam, Chandrayee Basu, Chengyuan Ma) — May 2025 – Aug 2025
Worked on error recovery in conversational systems with deployment constraints [12]

University of Illinois Urbana-Champaign | Conversational AI Lab *Urbana, IL, USA*
Graduate Research Assistant (Advisor: Dilek Hakkani-Tür) — Aug 2024 – Present

Korea University | Data Science & Business Analytics Lab *Seoul, South Korea*
Research Associate (Advisor: Pilsung Kang) — Apr 2024 – Jul 2024
Worked on retrieval-augmented language models specialized in Korean digital media

LG AI Research | Advanced Machine Learning Lab *Seoul, South Korea*
Research Scientist Intern (Mentors: Kyungjae Lee, Moontae Lee) — Apr 2023 – Mar 2024
Worked on improving long-form retrieval-augmented generation via planning [8]

NAVER AI Lab | Language Research Team *Seongnam, South Korea*
Research Scientist Intern (Mentor: Sungdong Kim) — Jul 2022 – Jan 2023
Worked on revealing innate bias in task-oriented dialogue research via interactive user study [7]
Worked on constructing safety benchmark for sensitive questions [4] and social biases [5]

Honors and Awards

Conference Travel Grant (Registration + 1,600 USD) ICLR 2026 [12] — Apr 2026
CS PhD Fellowship University of Illinois Urbana-Champaign — Aug 2024 – May 2025
Best Paper Award ACL 2024 Workshop on NLP for Conversational AI [7] — Aug 2024
Best Student Feedback Award, Research Ideation Competition Korea University — Feb 2021, Dec 2021
2nd Place (2,500 USD), MyData Acting Hackathon Ministry of Science and ICT — Oct 2019
Merit-based Scholarship, Computer Science Coursework Sungkyunkwan University — Spring 2018, Fall 2018

Publications

*, † = equal contribution

Published Papers

[17] **DuplexGen: Adaptive Synthesis of Human–AI Turn-Taking Dialogues**
Takyoung Kim*, Kang-wook Kim*, Sang Hoon Woo, Julia Hirschberg, Gunhee Kim, Dilek Hakkani-Tür
Conference on Empirical Methods in Natural Language Processing (Oral Presentation) EMNLP 2026

[16] **MMJail: Compounded Multilingual and Multi-Turn Vulnerabilities in LLM Bias Resistance**
Anisha Rao, Takyoung Kim, Dilek Hakkani-Tür
Workshop on Online Abuse and Harms WOAH, EMNLP 2026

[15] **Accurate Plate Reverb Parameter Estimation Using Two-Stage Evolutionary Search**
Byunghoo Park*, Jayeon Yi*, Takyoung Kim, Minje Kim
International Conference on Digital Audio Effects (Parameter Estimation Challenge Track, 2nd Place) DAFx 2026

[14] **AURA: A Diagnostic Framework for Tracking User Satisfaction of Interactive Planning Agents**
Takyoung Kim*, Janvijay Singh*, Shuhaib Mehri*, Emre Can Acikgoz, Sagnik Mukherjee, Nimet Beyza Bozdag, Sumuk Shashidhar, Gokhan Tur, Dilek Hakkani-Tür
IEEE Transactions on Audio, Speech and Language Processing (Q1, Impact Factor: 5.1) IEEE TASLP 2026
Workshop on Multi-Turn Interactions in Large Language Models MTI-LLM, NeurIPS 2025

[13] **From Documents to Segments: A Contextual Reformulation for Topic Assignment**
Hoonsang Yoon*, Takyoung Kim*, Wonkee Lee, Ilmin Cho, Dilek Hakkani-Tür†, Stanley Jungkyu Choi†
Annual Meeting of the Association for Computational Linguistics Findings of ACL 2026

[12] **ReIn: Conversational Error Recovery with Reasoning Inception**
Takyoung Kim, Jinseok Nam, Chandrayee Basu, Xing Fan, Chengyuan Ma, Heng Ji, Gokhan Tur, Dilek Hakkani-Tür
International Conference on Learning Representations ICLR 2026

[11] **Goal Alignment in LLM-Based User Simulators for Conversational AI**
Shuhaib Mehri, Xiaocheng Yang, Takyoung Kim, Gokhan Tur, Shikib Mehri, Dilek Hakkani-Tür
Transactions of the Association for Computational Linguistics (Oral Presentation at ACL 2026) TACL 2026

[10] **TD-EVAL: Revisiting Task-Oriented Dialogue Evaluation by Combining Turn-Level Precision with Dialogue-Level Comparisons**
Emre Can Acikgoz*, Carl Guo*, Suvodip Dey*, Akul Datta, Takyoung Kim, Gokhan Tur, Dilek Hakkani-Tür
Annual Meeting of the Special Interest Group on Discourse and Dialogue (Oral Presentation) SIGDIAL 2025

[9] **Premise-Augmented Reasoning Chains Improve Error Identification in Math Reasoning with LLMs**
Sagnik Mukherjee*, Abhinav Chinta*, Takyoung Kim, Tarun Anoop Sharma, Dilek Hakkani-Tür
International Conference on Machine Learning ICML 2025

[8] **Learning to Explore and Select for Coverage-Conditioned Retrieval-Augmented Generation**
Takyoung Kim, Kyungjae Lee, Young Rok Jang, Ji Yong Cho, Gangwoo Kim, Minseok Cho, Moontae Lee
Nations of the Americas Chapter of the Association for Computational Linguistics Findings of NAACL 2025

[7] **Revealing User Familiarity Bias in Task-Oriented Dialogue via Interactive Evaluation**
Takyoung Kim, Jamin Shin, Young-Ho Kim, Sanghwan Bae, Sungdong Kim
Workshop on NLP for Conversational AI (Oral Presentation, Best Paper Award) NLP4ConvAI, ACL 2024

[6] **DSTEA: Improving Dialogue State Tracking via Entity Adaptive Pre-training**
Yukyung Lee, Takyoung Kim, Hoonsang Yoon, Pilsung Kang, Junseong Bang, Misuk Kim
Knowledge-Based Systems (Q1, Impact Factor: 8.8) KBS 2024
Workshop on Knowledge Augmented Methods for NLP KnowledgeNLP, KDD 2023

[5] **KoSBi: A Dataset for Mitigating Social Bias Risks Towards Safer Large Language Model Applications**
Hwaran Lee*, Seokhee Hong*, Joonsuk Park, Takyoung Kim, Gunhee Kim, Jung-Woo Ha
Annual Meeting of the Association for Computational Linguistics (Industry Track) ACL 2023

[4] **SQuARe: A Large-Scale Dataset of Sensitive Questions and Acceptable Responses Created Through Human-Machine Collaboration**
Hwaran Lee*, Seokhee Hong*, Joonsuk Park, Takyoung Kim, Meeyoung Cha, Yejin Choi, Byoungpil Kim, Gunhee Kim, Eun-Ju Lee, Yong Lim, Alice Oh, Sangchul Park, Jung-Woo Ha
Annual Meeting of the Association for Computational Linguistics (Oral Presentation, Best Paper Nomination) ACL 2023

[3] **Oh My Mistake!: Toward Realistic Dialogue State Tracking including Turnback Utterances**
Takyoung Kim, Yukyung Lee, Hoonsang Yoon, Pilsung Kang, Junseong Bang, Misuk Kim
Towards Semi-Supervised and Reinforced Task-Oriented Dialog Systems SereTOD, EMNLP 2022

[2] **Mismatch between Multi-turn Dialogue and its Evaluation Metric in Dialogue State Tracking**
Takyoung Kim, Hoonsang Yoon, Yukyung Lee, Pilsung Kang, Misuk Kim
Annual Meeting of the Association for Computational Linguistics (Short) ACL 2022

[1] **Building an Integrated Framework of Korean Text Summarization and Text-to-Speech**
Takyoung Kim, Jina Kim, Hyeongwon Kang, Subin Kim, Pilsung Kang
Journal of Korean Institute of Industrial Engineers JKIIE 2022

Presentations and Talks

DuplexGen: Adaptive Synthesis of Human–AI Turn-Taking Dialogues
Midwest Speech and Language Days (Urbana, IL, USA) Apr 2026

Seamless Spoken Interaction with AI Companions: Naturalness and Efficiency
PhD Qualifying Exam (Subarea: AI, Committees: Minje Kim, Nancy Amato, Tong Zhang) Feb 2026

Data-Driven Approaches to Personalizing Spoken Interaction in Conversational AI
Seoul National University (Seoul, South Korea), Host: Pilsung Kang Dec 2025
Hanyang University (Seoul, South Korea), Host: Misuk Kim Dec 2025

Explainable and Efficient Turn-Taking Strategies for Full-Duplex Conversational AI
TTIC Summer Workshop on Foundations of Speech and Audio Foundation Models (Chicago, IL, USA) Sep 2025

Embodied Multi-Agent Planning in Conflicting Scenarios
Midwest Speech and Language Days (Notre Dame, IN, USA) Apr 2025

Relaxing Unrealistic Assumptions of Dataset and Evaluation Metric in Dialogue State Tracking
Master's Thesis (Committees: Pilsung Kang, Seoung Bum Kim, Sung Won Han) Jun 2022

Teaching Experience

CS546 Advanced Topics in NLP University of Illinois Urbana-Champaign Fall 2026
Teaching Assistant (Instructor: Dilek Hakkani-Tür)

CS598 Conversational AI University of Illinois Urbana-Champaign Spring 2026
Teaching Assistant (Instructor: Dilek Hakkani-Tür)

T02459101 Natural Language Processing Hankuk University of Foreign Studies Spring 2024
Teaching Assistant (Instructor: Yukyung Lee)

Industrial Machine Learning LG Electronics Spring 2022
Head Teaching Assistant (Instructor: Pilsung Kang)

Research Mentoring Hyundai Mobis Summer 2021
Research Project Assistant (Instructor: Pilsung Kang)

Industrial Machine Learning and Deep Learning LG Innotek, SK Hynix Summer 2021
Teaching Assistant (Instructor: Pilsung Kang)

Academic Service

Conference Reviewer

Annual Meeting of the Special Interest Group on Discourse and Dialogue (SIGDIAL)	’26
IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)	’26
International Conference on Learning Representations (ICLR)	’26’27
Asia-Pacific Chapter of the Association for Computational Linguistics (AACL)	’25
Conference on Language Modeling (COLM)	’25’26
Annual Meeting of the Association for Computational Linguistics (ACL)	’25’26
Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)	’25
European Chapter of the Association for Computational Linguistics (EACL)	’24’25’26
Empirical Methods in Natural Language Processing (EMNLP)	’23’24’25’26
Neural Information Processing Systems (NeurIPS)	’22’23’24’25’26

Journal Reviewer

IEEE Transactions on Audio, Speech, and Language Processing	’26
Knowledge-Based Systems	’23’24

Workshop Reviewer

Workshop on Online Abuse and Harms @ EMNLP	’26
Workshop on Multi-Turn Interactions in Large Language Models @ NeurIPS	’25

Organizer

Midwest Speech and Language Days	’26

Student Volunteer (On-Site)

Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)	’25